

Ardie Barry Sailis

PhD Candidate in Pharmaceutical Sciences (Health), Department of Pharmaceutical Life Sciences, Faculty of Pharmacy, Universiti Malaya.

Contact Information

Name	Ardie Barry Sailis
Professional Title	PhD Candidate in Pharmaceutical Sciences (Health)
Email	ardiebarrys@gmail.com
Affiliation	Department of Pharmaceutical Life Sciences, Faculty of Pharmacy, Universiti Malaya
Location	Kuala Lumpur 50603, Malaysia

Professional Summary

PhD candidate in Pharmaceutical Sciences (Health) at Universiti Malaya, specializing in tobacco and e-cigarette toxicology, pharmacology, and molecular systems biology. My doctoral research examines the endocrine, reproductive, and hematological effects of cigarette smoking, e-cigarette use, and dual use in Malaysian cohorts, with emphasis on miRNA-mediated testosterone regulation, mitochondrial dysfunction, oxidative stress, and blood cell abnormalities.

My broader research integrates molecular, cellular, and systems-level approaches to investigate how inhaled toxicants disrupt redox signaling, gene regulation, intercellular communication, and reproductive health. I also develop conceptual work on cellular signaling as dynamic regulatory circuits, applying systems and control-theory perspectives to reinterpret canonical signaling pathways as temporally regulated biological networks.

Experience

DEC 2023 - ONGOING

PhD Candidate / Researcher

Department of Pharmaceutical Life Sciences, Faculty of Pharmacy, Universiti Malaya, Kuala Lumpur, Malaysia

Research focus: e-cigarette and tobacco toxicology, reproductive health, testosterone regulation, miRNA biology, mitochondrial dysfunction, oxidative stress, and hematological effects.

- Conducting doctoral research on smokers, e-cigarette users, and dual users in Malaysia.
- Investigating miRNA-mediated testosterone regulation and blood cell abnormalities.
- Developing publication projects on e-cigarette toxicology, reproductive health, and systems biology.
- Writing narrative reviews, systematic reviews, and conceptual research articles.

2025 - ONGOING

Peer Reviewer

Provides peer-review support for scholarly manuscripts in toxicology, pharmacology, reproductive health, public health, molecular biology, and environmental health.

Reviewed manuscripts for American Journal of Preventive Medicine, Journal of Hazardous Materials Advances, Pharmacological Reviews, Progress in Biophysics and Molecular Biology, Toxicology Reports, PLOS One, and International Journal of General Medicine.

OCT 2022 - JAN 2023

Intern

Completed a full-time internship at Indah Water Konsortium Sdn Bhd, gaining hands-on exposure to laboratory operations, microbiology-related procedures, water quality assessment, analytical workflows, and research support activities.

Assisted with sample handling, test preparation, equipment use, documentation, and technical reporting while following laboratory safety practices and standard operating procedures within a professional environmental services setting.

Education

Doctor of Philosophy in Pharmacy - Health

Universiti Malaya, Kuala Lumpur, Malaysia

Bachelor of Science in Microbiology

With Distinction

Universiti Malaya, Kuala Lumpur, Malaysia

Science, Biology

Labuan Matriculation College, Malaysia

Publications

Accepted / In Press

- Ferroptosis in e-cigarette aerosol-associated respiratory injury.

Peer-Reviewed Publications

- E-cigarettes and erectile dysfunction: biological mechanisms and research challenges. International Journal of Impotence Research, 2026.
- Adipose as a Driver, Not a Bystander: A Modern Synthesis of Obesity-Related Erectile Dysfunction. Diabetes, Obesity and Metabolism, 2026.
- E-cigarette aerosols as systemic metabolic disruptors: integrated mitochondrial, circadian, and neurobehavioral mechanisms. Toxicology Mechanisms and Methods, 2026.

- CYP1A1 as a conserved metabolic circuit linking environmental sensing to immune regulation. Archives of Toxicology, 2026.
- Effects of secondhand exposure to e-cigarette aerosol on lung health: a systematic review. Journal of Public Health, 2026.
- MicroRNA-mediated disruption of testosterone signaling associated with e-cigarette exposure. Environmental Toxicology and Pharmacology, 2026.
- NRF2-KEAP1 as a redox signal-resolution circuit: Beyond the antioxidant switch. Progress in Biophysics and Molecular Biology, 2026.
- E-cigarette aerosol constituents modulate Leydig cell steroidogenic pathways: Evidence from experimental models. Molecular and Cellular Endocrinology, 2026.
- Mitochondrial dysfunction induced by E-cigarettes. Toxicology, 2025.

Skills

Research Areas

E-cigarette toxicology

Tobacco toxicology

Reproductive health

Molecular systems biology

Pharmacology

Toxicogenomics

Laboratory and Analytical Skills

Molecular biology

qPCR

Transmission Electron Microscopy

SPSS

Bioinformatics

Data interpretation

Literature synthesis

Systematic review methods

Mechanistic toxicology

Scholarly Skills

Scientific writing

Peer review

Research conceptualization

Manuscript preparation

Languages

English

Professional working proficiency

Malay

Native or bilingual proficiency

Certificates

Workshops

- ic-PHIR Metabolic Workshop, UiTM Puncak Alam (2023)
- Quantitative PCR (qPCR) Workshop, Vazyme, University of Malaya (2024)
- UV-Visible Double Beam Spectrophotometer (Shimadzu) Workshop (2025)
- Systematic Literature Review and Meta-Analysis Workshop (2025)

Conferences

- ic-PHIR Conference at UiTM Puncak Alam (2023)

- Sailis AB, et al. "Sexual Desire and Nicotine Dependence Among Smokers, E-Cigarette Users, and Dual Users in Malaysia: A Comparative Study" ISAM Regional Meeting 2024 Conference - Poster Presentation
- Early Career Researchers-Postgraduates Research Symposium (2024)

Projects

Doctoral Research Project

E-cigarette toxicology and reproductive health, with focus on sexual function, testosterone signaling, Leydig cell steroidogenesis, mitochondrial dysfunction, and secondhand aerosol exposure.

Cellular Signaling as Dynamic Regulatory Circuits

Conceptual framework interpreting canonical signaling pathways as dynamic regulatory systems with attention to feedback, signal duration, stress integration, and resolution failure.

E-Cigarette Toxicology and Reproductive Health

Research and writing project integrating mechanistic evidence on endocrine, vascular, inflammatory, and mitochondrial pathways affected by e-cigarette exposure.

Secondhand E-Cigarette Aerosol and Lung Health

Systematic review project assessing respiratory outcomes associated with passive exposure to e-cigarette aerosols.

References

Available upon request.